

WASP-12b

R-0540

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_181205 · created 2026-07-22T00:06:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458457.73216 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.029
Transit depth (Rp/Rs)^2	1.51 +/- 0.71 [%]
Orbital Inclination (inc)	76.83 +/- 2.46
Airmass coefficient 1 (a1)	1.033 +/- 0.024
Airmass coefficient 2 (a2)	-0.0245 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	#2 - [381, 207]
Optimal Aperture	3.98
Optimal Annulus	12.6
Transit Duration (day)	0.104 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.029 vs 0.1178 (deviation 0.15σ)
Duration fit vs published	0.104 d vs 0.125 d (deviation 1.39σ)
Mid-time O–C	-1.7 min
Depth SNR	3.7
Residual scatter	2.29 %

Light curve

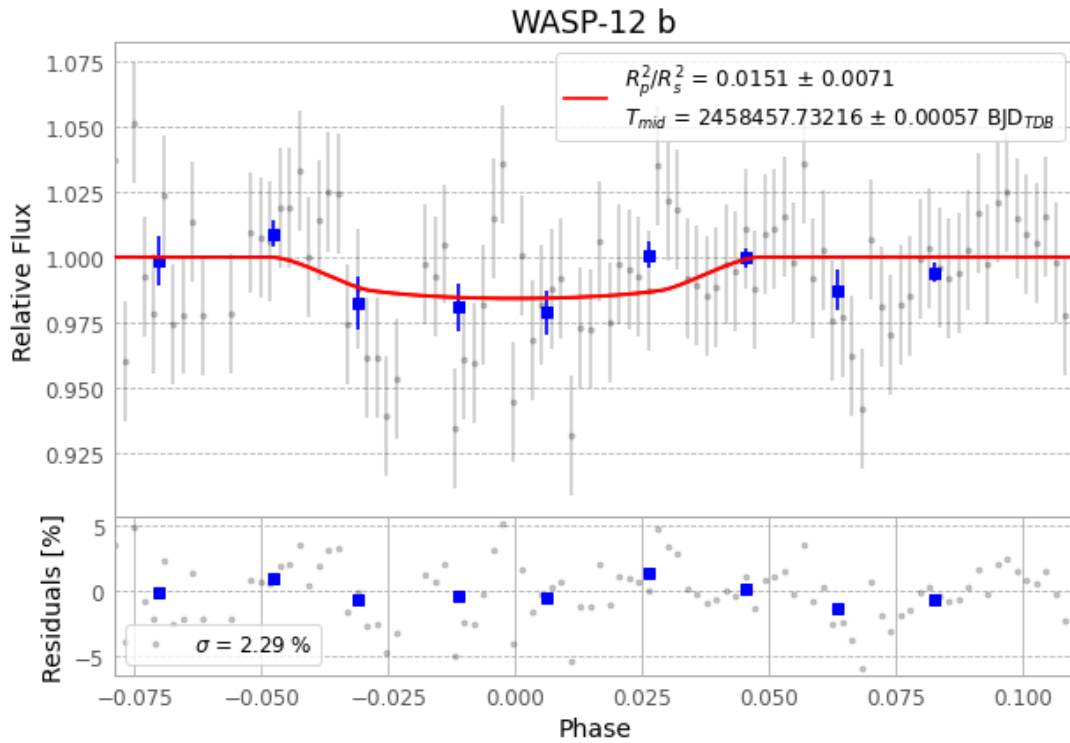


Figure 1 — FinalLightCurve_WASP-12 b_2018-12-04.png (SHA-256 0ff9a4159a48acb8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 127], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 32f080151bfd202e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

101 FITS frames (66 MB), WASP-12181205031812.FITS → WASP-12181205081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181205.log (56e2045a72fe...), FinalLightCurve_WASP-12 b_2018-12-04.png (0ff9a4159a48...), FinalLightCurve_WASP-12 b_2018-12-04.csv (634243be2813...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 00:06Z.